

**Institute of Computational Mathematics and Mathematical  
Geophysics of Siberian Branch of Russian Academy of Sciences  
Moscow Center of Fundamental and Applied Mathematics**

**holds**

**7th Workshop on Numerical Modeling  
in MHD and Plasma Physics:  
Methods, Tools, and Outcomes**

**Novosibirsk, Russia  
October 15-17, 2024**

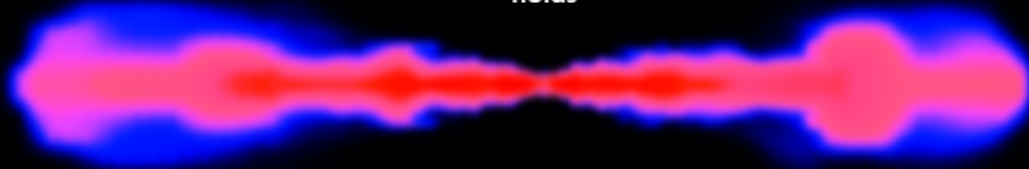
***BOOK OF ABSTRACTS***

# The conference poster

Institute of Computational Mathematics and Mathematical Geophysics  
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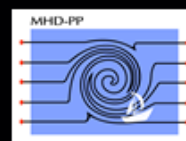
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7th Workshop on Numerical Modeling in MHD and Plasma Physics:  
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## Topics

Numerical methods for MHD equations  
Particle-in-cell method  
High performance computing  
Computational Fluid Dynamics  
Computational Physics of Plasma  
Computational Astrophysics

## Proceedings

Lobachevskii Journal of Mathematics

## Important dates

|                            |              |
|----------------------------|--------------|
| Abstracts deadline         | July, 25     |
| Notification of acceptance | August, 1    |
| Papers deadline            | September, 1 |
| Registration fee deadline  | October, 1   |

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## **Numerical modeling of gamma-ray burst spectra**

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The formation of gamma radiation resulting from a supernova explosion is a complex phenomenon and is not known precisely. In this work, we propose a model of gamma-ray burst from magnetorotational supernova after its explosion. We consider the Compton plasma cloud formed around the metallic core and the channel inside it with the collimated jet inside. The observed spectrum is proposed to be the result of rising from the star and cloud photons scattering during the relativistic stream acceleration. These phenomena are described by a three-dimensional model of relativistic radiative hydrodynamics including the full radiative transfer equation with scattering integral. Due to relativistic speed of the flow one needs to use a Compton scattering model that takes into the change in photon frequency when colliding with fast-moving electrons. To obtain numerical solution of the radiation gas dynamics equations the splitting method is used. So, for the radiative transfer equation with scattering the short characteristics method is used, and for the relativistic hydrodynamics the Godunov-type one. The algorithm is designed for tetrahedral meshes and adapted for computations on cluster systems with graphics processing units. During the simulations the plasma acceleration plasma in a conical channel under the influence of radiation during a gamma-ray burst was considered. The central object radiation pressure accelerated the matter up to 90% of the speed of light and the observed spectrum shifted from black-body radiation to the X-ray range. The spectra for observer different points of view are calculated.

## **Radiative transfer on nested meshes**

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Radiation plays a key role in astrophysical processes, and in particular, in star and planet formation. Including transfer of radiation in a numerical hydrodynamics code often requires solving systems of non-linear algebraic equations with respect to the radiation energy density and temperature of matter, in addition to a hydrodynamic equation describing the energy balance in the system based on the second law of thermodynamics. In my talk, I will describe, step by step, a transition from a simplified approach based on the polytropic equation of state to a more accurate scheme that considers full radiative transfer in the system. This will be done using the ngFEOSAD code, which is a numerical hydrodynamics code on nested meshes developed to simulate the initial stages of star and planet formation.

## Numerical modeling of charged particle dynamics in a helical magnetic trap

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A new method of an efficient plasma flow suppression in a helical magnetic field was recently proposed [1]. In the limit of small Larmor radius the particle Hamiltonian in such a field is fully integrable and the collisionless particle dynamics can be studied by methods of Hamilton dynamics [2]. In this work we present a numerical algorithm for calculating the magnetic field of a helical magnetic trap and investigate the collisionless motion of charged particles in the resulting field numerically. An important feature of the proposed algorithm is that it automatically fulfills the finite-difference conditions  $\text{div } \mathbf{B}=0$  and  $\text{div } \mathbf{A}=0$ .

Authors acknowledge support by Russian Science Foundation within the project № 24-21-00137.

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- [2] I.S. Chernoshtanov, D.A. Ayupov, Collisionless particle dynamics in trap sections with helical corrugation, Phys of Plasmas 28, 032502 (2021)  
<https://doi.org/10.1063/5.0040715>

## **Intense terahertz radiation from gaseous targets ionized by tailored two-color infrared laser beams**

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We study numerically, using particle-in-cell simulations with field ionization included, the excitation of electron currents in the interaction of a short intense two-color infrared laser beam with a transparent sub-millimeter gaseous target. The laser field consists of a strong circularly polarized fundamental component and its second harmonic of lower intensity. Our simulations demonstrate that for a sufficiently small interaction volume the plasma oscillations excited by an asymmetric ionization are almost spatially homogeneous and result in an efficient conversion of the electron energy into the energy of the emitted terahertz radiation. Moreover, it is shown that strong quasi-static electric field is present in the plasma for several hundred femtoseconds after the end of the ionizing two-color laser pulse.

Authors acknowledge support by Russian Science Foundation within the project № 24-21-00037.

## **Coarray Fortran Adaptive Mesh Refinement Code for Numerical Modeling of Relativistic Jets**

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The Relativistic jets are main sources of radio emission in the Universe. Multiscale flows in jets require of using to adaptive mesh technique. In talk will propose original the Patch-Block-Structured Adaptive-Mesh-Refinement technique for multi-scale modeling of relativistic jets. To use this technique, the numerical method was redesigned in a special way by means Coarray Fortran technology. On problem of jet evolution in interstellar medium, the applicability of the developed approach is shown.

This work was supported by the Russian Science Foundation (project 23-11-00014).

## **The scalable computational system on base mobile devices for numerical modeling in astrophysics**

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The report will outline the architecture of the mathematical and software of a distributed computing system with open source software for smartphones based on Android OS, integrated into a common network infrastructure. Building exascale systems based on mobile devices is a complex scientific task, since such systems must be potentially infinitely scalable, fault-tolerant, and must allow a guaranteed assessment of scalability when solving a specific problem. The BOINC system is the most well-known infrastructure for personal voluntary computing. However, it also has a number of disadvantages that must be eliminated when building a distributed computing system based on mobile devices with the performance of an exaflops class. The proposed architecture makes it possible to take into account the reconfiguration of a network of computing devices and build a guaranteed scalability assessment for heterogeneous distributed mobile computing systems[1]. The proposed distributed computing system is based on the single program multiple data (SPMD) programming model and the MapReduce distributed computing model. This model is focused on massively distributed computing, when the original problem allows decomposition into thousands or more independent subtasks. A feature of the proposed architecture of a distributed computing system is the fact that independent subtasks require time costs comparable to the “lifetime” of a computing node [2]. The reliability of the constructed scalability assessment was confirmed experimentally for a small number of mobile devices and theoretically using a proven simulation model for exascale systems[3]. The work focused on the use of the developed computing system to solve the problems of the collision of white dwarfs and the explosion of a thermonuclear supernova.

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3. I.S. Ulyanichev, D.V. Wiens. Architecture of a distributed computing system based on mobile devices //journal “Problems of informatics”. 2024, № 1. P. 74-97. DOI: 10.24412/2073-0667-2024-1-74-97.

## **Mathematical techniques used to obtain global transonic flows around compact objects**

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Accretion flows around compact objects like black holes, neutron stars and white dwarfs are necessarily transonic in nature or in other words have sonic point (or points). Similarly, winds from these objects can also be transonic depending on the constants of motion and flow parameters of the system. Bondi in 1952 presented a methodology to solve for both these flows simultaneously, utilising the fact that these flows are governed by Euler equations and are only differentiated by boundary conditions. Additionally, transonic flows have maximum entropy and would be always preferred by nature. The first step of this method requires to find the sonic point, and then from the sonic point, the Euler equations are integrated in different directions (inwards, to reach the central object and outwards, to reach infinity) to obtain the full global solution. However, at the sonic point which is a critical point, the gradient of velocity is undefined. Employing L'Hospital's rule we can find the gradient of velocity. Now, we have two values of velocity gradient, one positive which gives the wind solution, while the negative value gives the accretion solution. Although simple, this methodology proved to be a foundation for obtaining transonic accretion and wind solutions and is called the 'sonic point analysis' method. This method is widely used till now, and has been improvised from time to time in order to incorporate different processes in the system. However, in the presence of dissipation, finding of sonic points is not trivial and we need to take recourse of an iterative technique to obtain sonic points, which we will discuss in this work. Also, from the sonic point, integration inwards (for both the roots of velocity gradient) is not possible using the traditional integration methods. Hence, we propose here a modified methodology to obtain full global transonic solutions for both accretion and winds.

## **Applying Darwin's approximation to model open magnetic traps**

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When numerically modeling the motion of plasma in an open magnetic trap, kinetic equations are used for each of the plasma components. To calculate the intensities of magnetic and electric fields Maxwell's equations are added. Using a completely kinetic model results in the need to use a very small time step. The first reason is the large difference in the masses of the ion and electron components. It leads to the fact that the code spends most of its time computing the motion of electrons. The second reason is that Maxwell's equations describe electromagnetic waves propagating at the speed of light. However, in the problem of modeling processes in a plasma trap, the main speed is the Alfvén speed, and the influence of electromagnetic waves is insignificant. As the result a small time step requires large computational costs. In order to eliminate these disadvantages, we used before a hybrid plasma model, where the ion component of the plasma was described by a kinetic equation, and the electronic component was modeled in the hydrodynamic approximation.

This report deals with the new method based on Darwin's approximation. Previously, we proposed a new method for solving the equations of particle motion, allowing us to simulate the motion of electrons with a large time step, that eliminates the limitation on the time step associated with the large difference in the masses of ions and electrons. To eliminate the limitation on the time step associated with the speed of light, it is proposed to use the Darwin approximation. In this approximation, the vortex part of the electric field intensity is neglected, leading to the impossibility of propagation of electromagnetic waves. In addition, the hyperbolic system of Maxwell's equations becomes a system of equations of elliptic type, i.e. the solution scheme becomes absolutely stable. The disadvantage of Darwin's approximation is that at each time step seven Poisson equations must be solved. However, the possibility to use a large time step compensates for this disadvantage. To speed up the solution of Poisson's equations, the fast Fourier transform is used. The report presents the first results of computational experiments.

The work was carried out with the financial support of the Russian Science Foundation (grant number 24-21-00137).

## **A new way to improve the approximation of first derivatives for high-order SPH: theory and practice for the linear transport equation**

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Smoothed Particle Hydrodynamics (SPH) is often considered as a method of approximating partial differential equations on moving nodes that are irregularly located in space. The main idea of SPH is to construct a smooth approximating function from a discrete set of nodes using a finite smooth function called kernel. The support of the kernel is smaller than the computational domain, but covers several neighbor nodes (particles). In the classical version of SPH, to compute the derivatives of an approximating function one needs to differentiate a smooth kernel. The paper proposes a new method for approximating first derivatives in SPH, based on the idea of computing finite differences of functions known in nodes instead of kernel differentiation.

We obtained dispersion relations both for linear one-dimensional transport equation and its SPH-approximations. From the dispersion relation we found continuum and numerical phase velocity. We found previously that numerical phase velocity error correlates well with approximation error. Therefore many numerical results can be predicted theoretically.

We found that the new method for calculating derivatives gives the same or higher order of approximation than the classical one with comparable computational costs. In addition, it was found that the actual error of the solution obtained by the new method turns out to be several times less than when using the classical method even with a coarse resolution. The price for increasing accuracy is the narrowing of the range of applicability of the new method. Namely, it can be used in the case when the distance from the particle to the nearest neighbor is more than 3 times less than the radius of the smoothing kernel. Finally we confirmed all theoretical results in practice, using a high order kernel for numerical simulations.

The study was found by the Russian Science Foundation grant number 23-11-00142.

## **Computer algebra library for generation of dispersion relations of continuum mechanics models**

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Wave processes (processes of signal transport separate from the transport of matter) are a phenomenon of nature that interests a great number of researchers and engineers. Mathematical and computer models of continuum mechanics allow to reproduce many wave processes in nature.

One of the basic and effective methods of studying waves is the construction of dispersion relations - algebraic equations linking the various characteristics of the wave. With the help of analysis of dispersion relations it is possible, for example, to study the behavior of waves and verify computer models.

The technique of constructing dispersion relations is simple enough to learn, but as the system of equations under consideration becomes more complex, the number of calculations performed increases significantly. This, in turn, requires a large amount of monotonous manual labor and increased attention from the researcher. However, despite the development of symbolic mathematics systems and the demand for solving the problem, this process has not been automated so far even for the simplest equations.

The authors of this paper formalized an algorithm for finding dispersion relations for quasilinear homogeneous systems of one-dimensional equations and their discrete finite-difference analogues. This algorithm was implemented as a library in the Python language using the SymPy symbolic computation package (reference: <https://test.pypi.org/project/waves-searching/>).

## **Free fall of a Jupiter-like planet onto a protoplanetary disk**

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The theory of the formation of planetary systems predicts ejection of bodies of different masses, including planets, from the system into interstellar space [1]. In addition, isolated objects weighing up to 6 Jupiter masses, like stars, can form in star formation regions due to gravitational collapse [2]. According to modern estimates, in our Galaxy, the number of free-floating planets with a mass of several Jupiter masses should exceed the number of Main Sequence stars by at least two times [3]. This means that the convergence of a star and a free-floating planet should be considered as not an exceptionally rare event. Free-floating planets can be captured by the gravitational field of another star and become its satellite, or fly by, changing the trajectory. If a star has a protoplanetary disk, a free-floating planet may collide with it. In this paper, the problem of the passage of a giant planet through the protoplanetary disk of a young star is considered. A gas dynamic simulation has been performed for a wide range of parameters of the orbit of the incoming planet. The possibility of observing such an event is analyzed on the light curves of stars whose disks are observed at a small angle to the eye of sight (type UX Ori), and on images of disks observed from the pole.

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**Comparison of the turbulent spectrum in the development  
of Richtmayer-Meshkov instability numerically calculated  
with different approaches**

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With a short-term lateral impact of a shock wave (from Supernova, for example) on a contact discontinuity between gases of different densities, numerous turbulent vortices arise --- Richtmyer--Meshkov instability. The current work compares the results of a numerical solution of this problem in a specific two-dimensional formulation using various gas dynamic codes (FVM and SPH approaches). Various formulations of initial conditions, including those close to astrophysical conditions, have been studied. A turbulent spectrum was constructed and analyzed for each calculation. Parallel computations were carried out on video cards and a computer with distributed memory. The study was supported by a grant from the Russian Science Foundation №23-11-00142.

## **Spectral solvers of Maxwell's equations in Particle-In-Cell codes: numerical schemes and parallel implementation**

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To solve Maxwell's equations, most particle-in-cell (PIC) simulation codes employ the Finite-Difference Time-Domain (FDTD) method, which is subject to numerical dispersion. In contrast, spectral solvers are free of numerical dispersion effects and provide a high-quality numerical solution. This paper discusses some issues encountered when integrating spectral solvers into PIC codes. We consider the problem of distributed calculations and analyze schemes for replacing the global Fast Fourier Transform (FFT) with local FFTs within a computational domain. Additionally, we describe various Perfectly Matched Layer (PML) boundary condition schemes adapted for spectral solvers and propose a new method for effectively generating an electromagnetic field at the boundary of a computational domain. Some of the schemes under consideration have been successfully implemented in the Picador code using the OpenMP and MPI technologies.

## **Genetic algorithm for searching unipolar and bipolar single-flux-quantum pulse sequences for two-qubit quantum operation**

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Nowadays most of superconducting quantum processors use charge qubits of a transmon type. Traditional schemes for implementing quantum operations on them are based on the transmission of analog microwave signals but they have difficulties with managing indices and reading qubits. A promising approach is the use of superconducting digital circuits operating with single-flux-quantum (SFQ) pulses. In our previous work we have shown how to use SFQ control sequence to implement the rotation gate around the Y axis by an angle  $\pi/2$  with fidelity over 99.99%. A next important step in further development is the implementation of two-qubit operations. Here we introduce a genetic algorithm for finding unipolar or bipolar SFQ control sequences that implement an original control protocol, using the example of the ZX operation. The algorithm also takes into account the control of leakage to higher energy states in the system. Its parallel implementation can find the appropriate sequence for arbitrary system parameters from a practical range in a reasonable time. As a result of numerical experiments, control sequences were found that implement the ZX operation with fidelity 99.9%.

## **Particle-in-cell simulation of plasma dynamics in an electric field**

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In a strongly non-isothermal plasma with hot electrons and cold ions ( $T_e \gg T_i$ ), low-frequency quasi-electrostatic oscillations can be excited, which are significantly influenced by ions. Under these conditions, the ion motion can be described hydrodynamically without taking into account the magnetic field and the electrons are in equilibrium, and their density is determined by using the Boltzmann formula [1]. In the long-wave part of their spectrum, the oscillations under consideration are characterized by a linear dispersion law and therefore these oscillations are called ion-acoustic. Their speed is determined by the electron temperature and the ion mass. In the short-wave part of the spectrum, the frequency of ion-acoustic oscillations is close to the ion Langmuir frequency. However, to study the influence of thermal motion of particles on low-frequency oscillations, the hydrodynamic approximation is insufficient and a kinetic description of the plasma is necessary [1], i.e., solving the Vlasov equations. The used particle-in-cell method is the most effective and universal method for solving the Vlasov equations [2].

The work studies the possibility of numerical modeling of such oscillations, as well as physical processes related to these oscillations, using the particle-in-cell method. For example, an important question is whether a model particle can be infinitely accelerated in a plasma or whether Debye screening will slow down the particles. The problem is solved in a one-dimensional formulation. The equation of conservation of total energy is used to control the accuracy of calculations.

The work have been developed in the frames of the budget project ICM&MG SB RAS 0251-2022-0005.

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## Impact interactions of protostellar clumps and the formation of jets and streamers arising from the collision of molecular clouds

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The paper presents the results of modeling the formation of protostellar structures that arise during supersonic collisions of molecular clouds. Most of these superdense structures are gravitationally bound and new stars are subsequently formed from them. Recent observations have shown that the interaction of such protostellar structures with each other plays a significant role in the formation of young stellar objects (YSOs) and binary star systems. In addition, observations show that the accretion phase is often accompanied by a powerful ejection of bipolar jets of matter.

The resulting motion is described by a system of three-dimensional Euler equations taking into account gravity [1]. The conducted modeling shows that after the formation of a protostar in a gravitationally unstable core, the accreting gas leads to the formation of a circumstellar disk, which is the source of jets (Fig. 1). Such jets, as well as the collisions of these protostars with each other, have a significant impact on the processes of star formation.

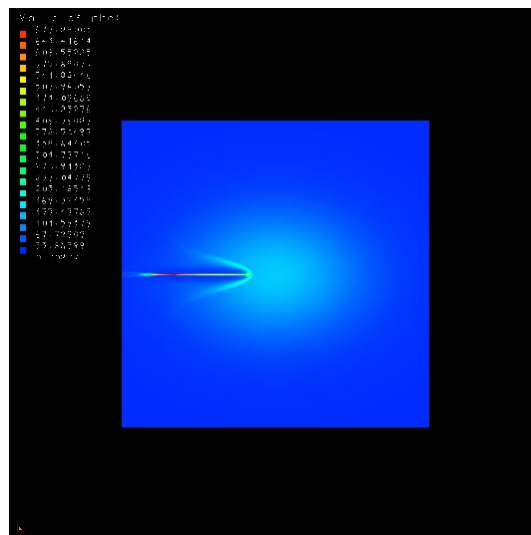


Fig. 1 Formation of a jet and its interaction with an elliptical molecular cloud

### Literature

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## **Neutron star kicks from magnetorotational supernovae**

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Rapidly moving neutron stars are believed to gain high linear velocities – kicks – in aspherical supernova explosions. The mechanism of the kick formation is probably connected with anisotropic neutrino flash, and/or anisotropic matter ejection during the explosion. In this paper, we investigate a neutron star kick origin in a magnetorotational (MR) supernova explosion model. The simulations have been done for series of core collapse supernova models of a massive star with rotation and initial equatorially asymmetric magnetic fields. We have conducted 2D MHD simulations, considering the protoneutron star kick and explosion properties in three different asymmetric magnetic field configurations. The simulations show, that in the MR supernova model protoneutron star kicks are formed with velocities up to  $\sim 500$  km/s during  $\sim 1$  second after the core bounce, due to asymmetric matter ejection in jets, what may explain the observed kick velocities.

## **Model of high-speed flows in the quasi-stationary plasma accelerator for the regime of electron current-transport**

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Model of flows has been developed to create powerful electric rocket engines (ERE) using the quasi-stationary plasma accelerator (QSPA). Analytical model of two-dimensional axisymmetric flows is based on the stationary equations of two-fluid magnetohydrodynamics taking into account the Hall effect for the ideally conducting plasma in the smooth channel approximation [1]. Earlier the experimental, theoretical and numerical studies in the QSPA have been carried out for the mixed and ion current-transport regimes, respectively, for solid and rod electrodes in the installations with the equipotential electrode surfaces (see, e.g., [2,3]). New idea of using non-equipotential electrodes in the channel of the QSPA is associated with the electron current-transport regime, which ensures the slipping of ions along anode and cathode, excluding the formation of large-scale layers associated with the heavy ion component of the plasma. At the same time, the electron current-transport regime provides a compact implementation of the ERE based on the QSPA for space applications. As a result of studying this regime, the nozzle-shaped channel, the transonic flows in the profiled channel, the potential distributions along non-equipotential electrodes, as well as the integral characteristics of the flow, including mass flux and thrust, were determined [4]. The potential distributions along non-equipotential electrodes found for the first time provide initial presentation of the implementation of this regime in practice.

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## **Magnetorotational processes in supernova.**

### **Explosion, MRI and symmetry violation.**

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Core-collapsed supernovae (CCSN) physical mechanism is one of the important and not completely solved astrophysical problem. Magnetorotational (MR) mechanism of supernova explosion (Bisnovaty-Kogan, 1970) explains many important features of the CCSN. The energy of explosion comes from the rotational (initially gravitational) energy. The magnetic field works here like transmission belt. Numerical simulations show that MR mechanism can give the CCSN explosion energy consistent with observations. The development of the MR explosion magnetorotational instability (MRI) is developing. The mechanism of the mirror symmetry violation in MR explosion could explain formation of the rapidly moving radiopulsars.

**Mathematical Model of Thermocurrents Based on Calculation  
of Electrical Resistance and Thermopower  
As an Integral over Electron Energy**

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Melt motion is one of the most destructive effects of instability developing in modern thermonuclear plasma facilities. As the material of the limiter or divertor walls melts and is further heated, they begin to evaporate. At the contact of the melt and the evaporated gas, melt motion is driven by thermoelectric effects due to the large magnetic field required for plasma confinement. The interaction of the plasma with the surface is a major problem when a divertor material is chosen in designing thermonuclear reactors. Possible high-energy neutron irradiation and high thermal loads impose severe limitations on the material properties. Expected high thermal stresses cause material erosion and melting. Rapid melting leads to changes in the divertor material surface caused by the motion of the melt. Such motion can be initiated by gravity, by the influence of the gas or plasma on the melting process, and by the Lorentz force. We consider a model of current distribution in a tungsten sample and a vapor layer produced when the surface is heated by an electron beam. The model is based on solving electrodynamic equations and a twophase Stefan problem in cylindrical coordinates. Based on the temperature distribution in the computational domain, the electrical resistance and thermopower are calculated via an integral over the electron energy at each grid node. The electromagnetic field configuration is a possible source of rotation of the substance, which is observed in experiments. The simulation results demonstrate the role of thermionic emission and the way of model development. The results of the simulation show that the choice of approximation of the material parameters have a great influence on the decision. The model parameters are taken from the experiments on the Beam of Electrons for materials Test Applications (BETA) stand, created at the BINP SB RAS.

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## **On positivity of the local iteration modified (LIM) time integration scheme**

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Local iteration modified (LIM) scheme is an explicit time integration method where Chebyshev polynomial-based iterations are employed to provide a stable and accurate numerical solution. The scheme is successfully applied for solving linear as well as nonlinear initial-value problems (IVPs) arising in various real-life applications. The LIM scheme essentially differs from an implicit scheme where linear system solution is done by Chebyshev iterations in two key aspects: (1) the LIM scheme is constructed to be more accurate for the important low frequency eigenmodes, (2) the number of Chebyshev iterations is chosen, for a given time step size, to have a stable time integration (rather than to solve the equation in an implicit scheme). This makes the LIM scheme quite efficient: for parabolic problems the number of steps in the LIM scheme is proportional to  $1/h$  with  $h$  being the space grid resolution. At the same time, the scheme is explicit, algorithmically simple and straightforward to parallelize.

In this work we thoroughly study another important property of LIM scheme, namely, its positivity: the scheme is known to preserve the non-negativity of the IVP solution. The positivity property is, as far as we know, unique for iterative schemes, as it is in general not clear how to guarantee positivity in the course of iterations. Up to now, positivity of the LIM scheme has been shown experimentally in numerical tests as well as, partially, theoretically. Our aim here is to provide further theoretical insight on the positivity of the LIM scheme. In particular, the stability function of the LIM scheme is studied and explicit expressions for the coefficients which determine the positivity are derived. Furthermore, some numerical test problems where positivity properties are crucial are considered.

## **Adaptive choice of a time step for numerical integrating parabolic equations with the explicit iterative Chebyshev scheme**

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The construction of the time step selection procedure for the LIM parabolic scheme is presented. This scheme has been designed for unsteady applications and presents an explicit iterative time integration solver which accuracy and stability for arbitrary large time steps are provided by special construction of Chebyshev polynomial. We propose to apply the LIM scheme to solve parabolic equations with the time step selection procedure. This procedure considers the dynamics of the solution and the time dependence of the equation entrees. A weakness of the procedure is difficulties of generalization to the multidimensional case. The LIM scheme is universal in terms of applicability to multidimensional discretization on arbitrary grids, including unstructured grids, which is relevant in a wide class of practical applications.

The complete algorithm is as follows. The predicted solution is found by the explicit scheme. A priori choice of a time step size is made by estimating the approximation error using the implicit scheme: the grid solution at the current layer obtained by the LIM scheme and the predicted solution are substituted into the implicit scheme formula as the solution at the lower and upper layers, respectively. The solution of the next time layer with the obtained predicted step is computed by the LIM scheme. The numerical results in the case of one-dimensional heat conduction equation demonstrate adaptation efficiency of the LIM scheme. Generalization of the approach to the multidimensional case is straightforward.

## Mathematical modeling of output plasma flow in SPT

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The paper proposes a mathematical model that allows calculating the output parameters of plasma in a stationary plasma thruster (SPT), bypassing the calculation of the main plasma processes in the thruster. It turns out that this is possible, provided that the output parameters are set along the SPT axis. Moreover, the proposed model allows taking into account the influence of wall conductivity [1], rotation of the side walls of the thruster around the common axis, and non-Spitzer conductivity of plasma in the SPT [2], which takes into account the contribution of the radial magnetic field to the conductivity, on the plasma output parameters. The determining factors underlying the proposed approach are the two-fluid nature of plasma, including full consideration of the electron inertia [3], and dissipative processes: magnetic and hydrodynamic viscosities of electrons and ions, thermal conductivity of plasma, and temperature relaxation. The simplest model is obtained under the assumption of plasma incompressibility. In this case, the mathematical calculation of the output diagrams is reduced to solving a boundary value problem for a certain linear system of 8th-order ODEs with variable coefficients on a segment. The report proposes two methods of successive approximations for the numerical solution of the obtained boundary value problem. The calculation results are presented, demonstrating the efficiency of the mathematical model.

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## **Mathematical aspects of the study of output plasma flow in SPT**

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A mathematical study of a special class of boundary value problems on a segment with Dirichlet-type boundary conditions for a linear system of ordinary differential equations (ODE) with variable coefficients is carried out. These problems arise when calculating the output parameters of plasma in stationary plasma thruster (SPT) used to correct the orbits of spacecraft. The plasma was assumed to be incompressible with constant coefficients of conductivity, hydrodynamic viscosity and thermal conductivity of electrons and ions, and temperature relaxation. The ODE system under study is of the eighth order and, in physically important cases, theorems of existence and uniqueness of a solution are proved for the posed boundary value problems, Green's functions are constructed, and two methods of successive approximations converging to a solution of the boundary value problem are proposed.

## **On the exact solution to the Vlasov-Poisson system in a bounded domain**

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We consider the Vlasov-Poisson system with nontrivial magnetic field of a special structure in a bounded domain. This system is considered with respect to three unknown functions: the potential of a self-consistent electric field and two distribution functions of charged particles (positively charged ions and electrons).

A large number of papers have been devoted to the existence of solutions to the mixed problems for the Vlasov-Poisson system. In [1]-[3] special attention is paid to the following factors: the model of a two-component plasma is considered, the influence of a non-trivial external magnetic field is taken into account, the distribution functions are compactly supported (or compactly supported in cross-section). The existence of stationary solutions to the first mixed problem with a nontrivial potential and the structure of the corresponding distribution functions are investigated in [2]. However, it is difficult to use these results directly as the exact solutions for numerical methods, since for a test solution it is necessary not only to explicitly specify the distribution functions, but also to determine the explicit form of the electric field potential function. We construct new exact stationary solution of the first mixed problem for the Vlasov-Poisson system in a bounded domain. The constructed solution consists of three functions written explicitly, and the magnetic field has a special non-uniform structure.

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## **Computational model of the composite material heating process when its surface is exposed to a laser**

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Modern plasma confinement systems assume contact between peripheral plasma and reactor walls. At the moment, tungsten is used as contacting material of the thermonuclear reactor and some of the components for the tokamak. The latest studies have shown that under the influence of high-density plasma irradiation we observe tungsten cracks, intense erosion, macroscopic particle emission, melting at relatively small temperatures and other processes that, as the result, can lead to the rapid breakdown of diverter's tungsten tiles [1]. High-temperature ceramic is promising alternative material to use as shield coating for plasma components. One of the main advantages of the ceramic compared to the other materials is its high resistance to intense thermal loads [2]. To protect tungsten plates from plasma exposure and, thus prevent the development of the above processes, we can use a renewable boron carbide ( $B_4C$ ) coating, which has a high melting temperature. In experiments at the BETA facility (Budker Institute of Nuclear Physics), the effect of laser radiation on samples of rolled tungsten and a boron carbide-tungsten composite were studied. The heat from the laser beam, absorbed by the surface of the sample, spreads deep into the sample, the maximum temperature is reached at the center and decreases toward the edges. The report presents a new model for heating a sample of a boron carbide-tungsten composite when the surface is heated by a laser. The model is based on the solution of the heat equation in an axially symmetric formulation [3]. The modeling domain is a cross-section of the sample. The numerical implementation is based on the Douglas-Rachford scheme and the tridiagonal matrix algorithm. The calculation results showed that boron carbide, having a higher heat capacity compared to tungsten, heats up more, but due to the low thermal conductivity it slows down the spread of heat into the depths. The modeling results are in demand for the analysis of results and experiments planning.

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## **Development of test problems for realization with Vlasov Poisson equations by PIC method.**

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The Particle-in-Cell (PIC) method is effective numerical approach widely used for plasma dynamics modeling. The relevance of the Particle-in-Cell method is due to its high accuracy, versatility and computational efficiency. With the development of computing technology and the increasing power of supercomputers, the importance of the PIC method in scientific research and industrial applications continues to grow, contributing to new discoveries and technological advances in plasma technology.

Originally developed for modeling fluid flows, the particle-in-cell (PIC) method is popular for solving the Vlasov-Maxwell and Vlasov-Poisson systems of equations [1, 2]. PIC method approximates the distribution function by a set of computational particles that move in self-consistent electric and magnetic fields. The implementation algorithms of the PIC-method ensure conservation of energy, momentum and charge. To obtain adequate modeling results, it is necessary to use parallel algorithms [3]. The convergence of the PIC method has been proven. But the existence of numerical noise of the particle method complicates the testing of the algorithms. The problem of reconstructing continuous analytic distribution functions by finite samples of computational particles arises.

The aim of the work is to develop new tests for codes based on PIC-method on the basis of qualitative solutions on a compact medium. The developed solutions are close to the formulations of the problems of plasma confinement in multi-mirror trap [4], which gives a great advantage to the new set of tests.

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## **The effect of temperature on the result of mathematical modeling of plasma transfer in a helical magnetic field**

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The report presents the first results of refining the mathematical model of plasma transfer in a spiral open magnetic SMOLA trap created at the Budker Institute of Nuclear Physics SB RAS [1]. Plasma confinement in the device is carried out by transmitting a pulse from a magnetic field with helical symmetry to a rotating plasma. The mathematical model is based on the stationary plasma transfer equation [2]. The method of accounting for the influence of the coefficients of the model, taking into account additional information, is described. The dependence of temperature on coordinates is obtained, at which a qualitative correspondence of the calculation to experimental data is observed. As a consequence of the initial problem, ordinary differential equations are obtained, which are planned to be used to refine the coefficients. A mathematical model has been developed to predict the parameters of plasma confinement in designed devices with a spiral magnetic field

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